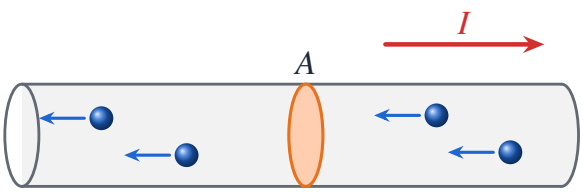
CURRENT ELECTRICITY

Current • Drift Velocity

JEE & NEET
PHYSICS

1 Electric Current

$$I = \frac{dq}{dt}, \qquad I_{\text{avg}} = \frac{\Delta q}{\Delta t}$$

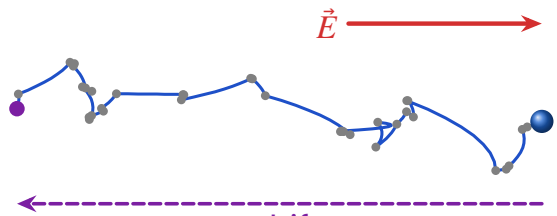


electrons move against I

Unit A = C s^{-1}. A scalar: currents add algebraically. Direction = flow of + charge.

2 Drift Velocity

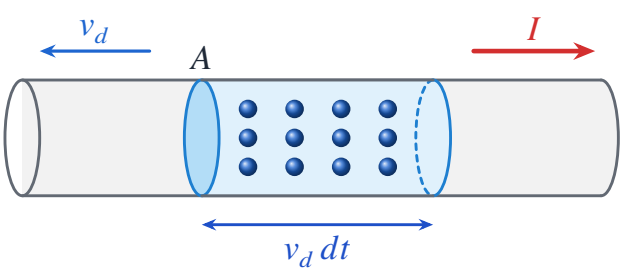
$$v_d = \frac{eE\tau}{m} \qquad (\vec{v}_d \text{ against } \vec{E})$$



τ = mean time between collisions. $v_d \sim 10^{-4}$ m/s; random thermal speed $\sim 10^5$ m/s.

3 Current from Drift

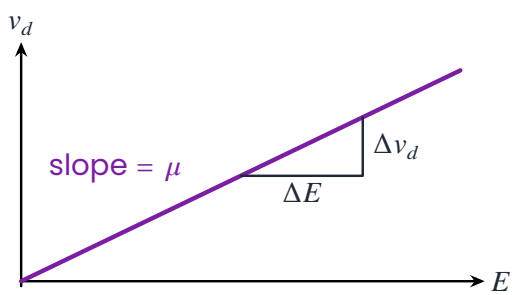
$$I = neAv_d$$



n = free electrons per m^3. Same I along a wire: where it is thinner, v_d is larger.

4 Mobility

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}$$



Unit $m^2\ V^{-1}\ s^{-1}$, taken positive. Conductivity $\sigma = ne\mu$.

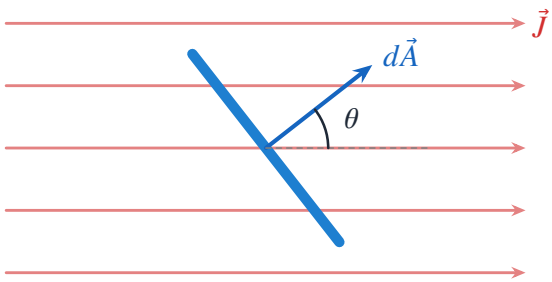
CURRENT ELECTRICITY

Current Density • Ohm's Law

JEE & NEET PHYSICS

5 Current Density

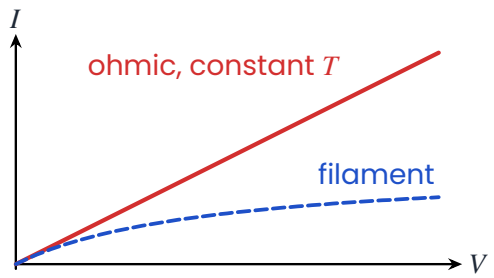
$$J = \frac{I}{A}, \qquad I = \int \vec{J} \cdot d\vec{A}$$



J uniform over an area $A \perp$ the flow; otherwise $I = \int \vec{J} \cdot d\vec{A}$.

6 Ohm's Law

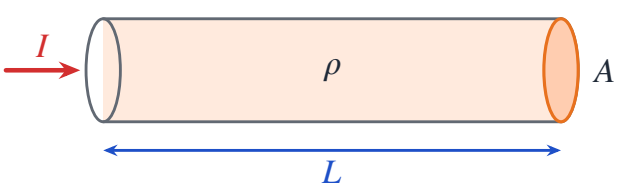
$$\vec{J} = \sigma \vec{E}, \qquad V = IR$$



Only for an ohmic conductor at constant temperature. $\sigma = \frac{ne^2\tau}{m} = \frac{1}{\rho}$.

7 Resistance of a Wire

$$R = \frac{\rho L}{A}$$



Uniform wire, current along its length. ρ depends on the material and T, not on the shape.

8 Quick Numerical: Drift Speed

A copper wire of cross-section 1.0 mm^2 carries 3.4 A; $n = 8.5 \times 10^{28}$ m^{-3}. Find v_d.

$$v_d = \frac{I}{neA} = \frac{3.4}{1.36 \times 10^4}$$

$$v_d = 2.5 \times 10^{-4} \text{ m/s}$$

A metre takes 67 min, yet a bulb lights at once.

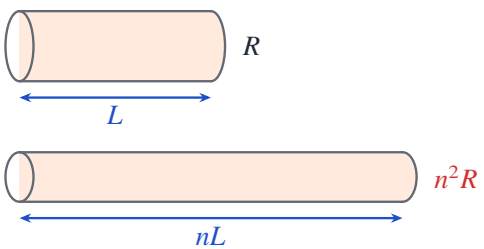
CURRENT ELECTRICITY

Resistivity • Colour Code • Series

JEE & NEET PHYSICS

9 Stretching a Wire

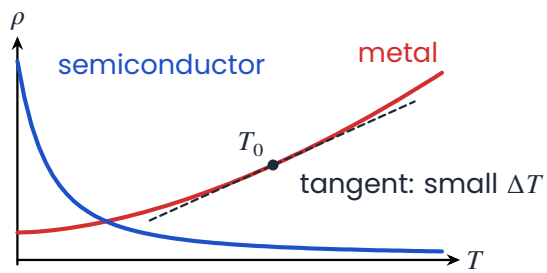
$$L \to nL: \quad R \to n^2 R$$
$$R \propto L^2 \propto r^{-4}$$



Volume and ρ stay constant. Radius halved: $R \to 16R$.

10 Resistivity and Temperature

$$\rho_T = \rho_0 [1 + \alpha (T - T_0)]$$



Metals, small ΔT only ($\alpha > 0$). Semiconductors: ρ falls as T rises. Manganin: tiny α.

11 Resistor Colour Code

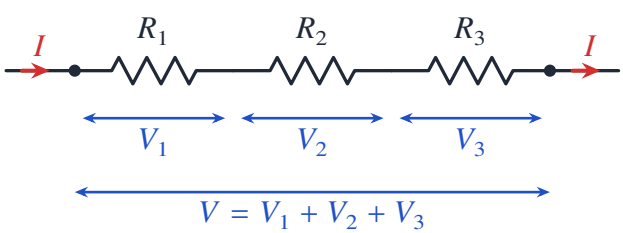


Colour	Digit	Colour	Digit
Black	0	Green	5
Brown	1	Blue	6
Red	2	Violet	7
Orange	3	Grey	8
Yellow	4	White	9
Gold	±5%	Silver	±10%

Yellow Violet Red Gold = 4.7 kΩ ± 5%

12 Resistors in Series

$$R_s = R_1 + R_2 + R_3$$



Same I through each; $V_i = IR_i$, so the biggest R takes the most V. $R_s >$ the largest.

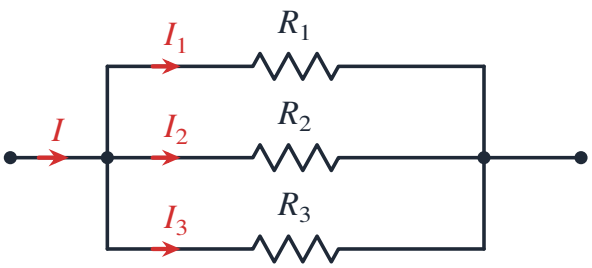
CURRENT ELECTRICITY

Parallel • EMF • Grouping of Cells

JEE & NEET PHYSICS

13 Resistors in Parallel

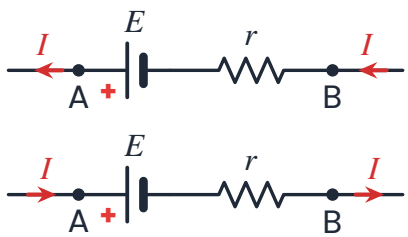
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$



Same V across each; $I_i = V/R_i$. $R_p <$ the smallest; for two, $R_1R_2/(R_1 + R_2)$.

14 EMF and Internal Resistance

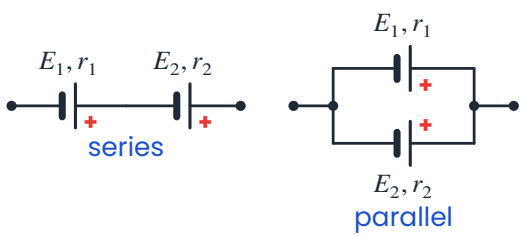
$$V = E - Ir \quad \text{(discharging)}$$
$$V = E + Ir \quad \text{(charging)}$$



Load R: $I = E/(R + r)$. Open: $V = E$. Short: $I = E/r$.

15 Cells in Series and Parallel

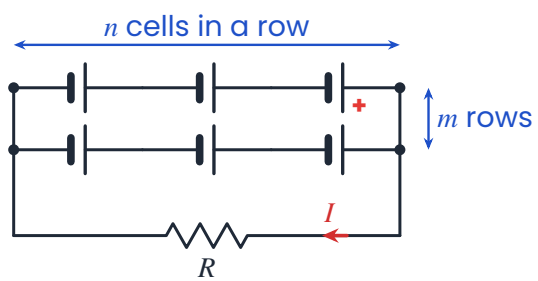
$$E_s = E_1 + E_2, \quad r_s = r_1 + r_2$$
$$E_p = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}, \quad r_p = \frac{r_1 r_2}{r_1 + r_2}$$



Series: + to − (reversed: $E_1 - E_2$). E_p: + to +.

16 Mixed Grouping

$$I = \frac{nE}{R + nr/m}$$



mn identical cells (E, r). Largest I when $R = nr/m$.

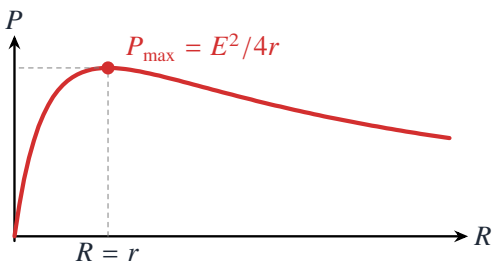
CURRENT ELECTRICITY

Maximum Power • Kirchhoff's Laws

JEE & NEET PHYSICS

17 Maximum Power Transfer

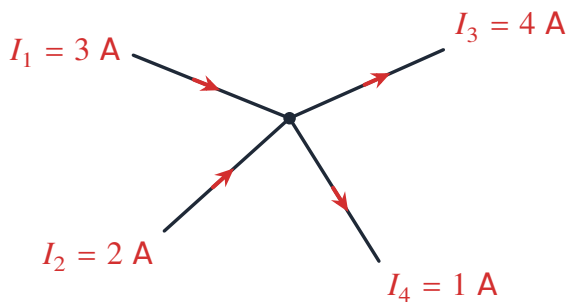
$$P = \frac{E^2 R}{(R+r)^2}, \qquad P_{\max} = \frac{E^2}{4r}$$



E, r fixed, R varied: peak at $R = r$, efficiency 50%.

18 Junction Rule

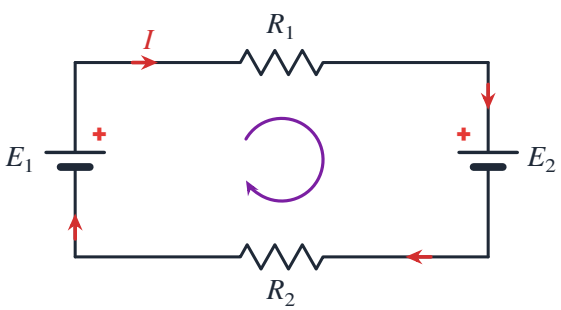
$$\sum I_{\text{in}} = \sum I_{\text{out}}$$



Conservation of charge: no charge piles up at a junction (steady state).

19 Loop Rule

$$E_1 - IR_1 - E_2 - IR_2 = 0$$



$\sum \Delta V = 0$ round a loop (energy).
Across R along I: $-IR$; cell − to +: $+E$.

20 Quick Numerical: Cell and Load

EMF 12 V, internal resistance 1 Ω, load 5 Ω. Find I, V, the load power, and the most power any load can take.

$$I = E/(R+r) = 2 \text{ A}$$
$$V = E - Ir = 10 \text{ V}$$
$$P = I^2 R = 20 \text{ W}$$

$R = r = 1\ \Omega$: $P_{\max} = \dfrac{E^2}{4r} = 36$ W

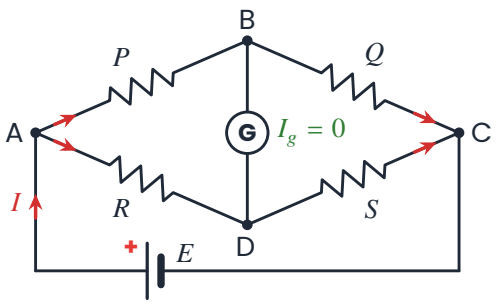
CURRENT ELECTRICITY

Wheatstone • Meter Bridge • Potentiometer

JEE & NEET
PHYSICS

21 Wheatstone Bridge

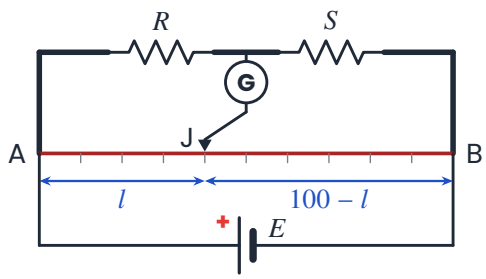
$$\frac{P}{Q} = \frac{R}{S} \quad \text{(balanced)}$$



Balanced: $I_g = 0$ (B, D equipotential).

22 Meter Bridge

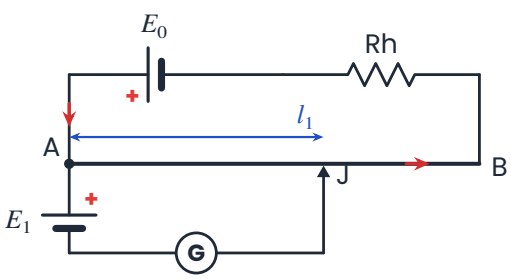
$$\frac{R}{S} = \frac{l}{100 - l} \quad (l \text{ in cm})$$



At the null point ($I_g = 0$); uniform wire; end corrections ignored.

23 Potentiometer: Compare EMFs

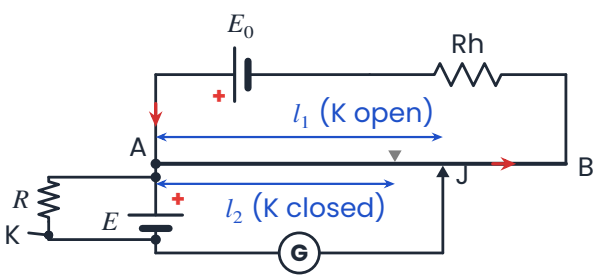
$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$



Uniform wire, steady current, $E_0 > E_1, E_2$; + ends at A.

24 Potentiometer: Find r

$$r = R\left(\frac{l_1}{l_2} - 1\right)$$



Uniform wire, same steady driver current for l_1 and l_2, $E_0 > E$.

CURRENT ELECTRICITY

Power • Heating • Bulbs

JEE & NEET PHYSICS

25 Electrical Power

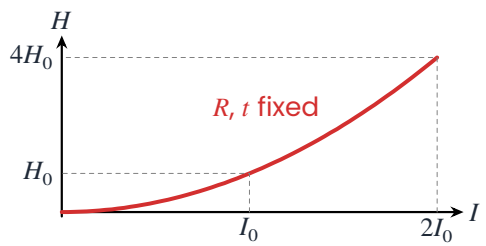
$$P = VI = I^2R = \frac{V^2}{R}$$

Given	Use	Then
any device	$P = VI$	always true
same I (series)	$P = I^2R$	$P \propto R$
same V (parallel)	$P = V^2/R$	$P \propto 1/R$

I^2R and V^2/R need $V = IR$ (a resistor). A motor or a charging cell: only VI.

26 Energy and Joule Heating

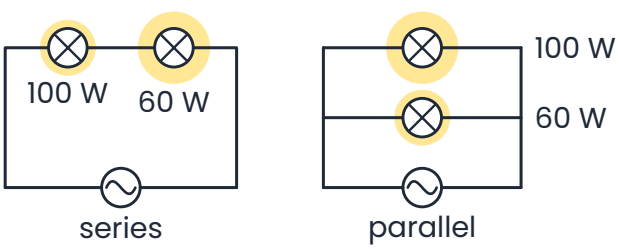
$$W = VIt, \qquad H = I^2Rt$$



Steady V and I; otherwise $W = \int VI\,dt$. $H = W$ only in a pure resistor.

27 Bulbs in Series and Parallel

$$R = \frac{V_{\text{rated}}^2}{P_{\text{rated}}}$$



Same rated V; R taken constant (filament heating ignored). Series: lower wattage brighter; parallel: higher.

28 Quick Numerical: Bulbs

100 W, 60 W bulbs (220 V) in series on 220 V, R constant. Which is brighter?

$$R_{100} = 484\ \Omega, \qquad R_{60} = 807\ \Omega$$

$$P_{100} = \frac{220^2 \times 484}{1291^2} = 14.1\ \text{W}$$

$$P_{60} = \frac{220^2 \times 807}{1291^2} = 23.4\ \text{W}$$

Same I, so $P \propto R$: the 60 W bulb.

CURRENT ELECTRICITY

Quick Revision • Traps

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29 Every Result and When It Holds

Result	Formula	Holds only if
Ohm's law	$V = IR,\ \ \vec{J} = \sigma\vec{E}$	ohmic conductor, constant T
Drift	$v_d = eE\tau/m,\ \ I = neAv_d$	τ = mean free time
Resistance	$R = \rho L/A$	uniform wire, I along L
Stretched wire	$R \propto L^2 \propto 1/r^4$	volume and ρ constant
ρ and T	$\rho = \rho_0(1 + \alpha\Delta T)$	metals, small ΔT
Terminal voltage	$E - Ir$; $E + Ir$	discharging ; charging
Cells in parallel	$E_p = (E_1r_2 + E_2r_1)/(r_1 + r_2)$	+ to +
Max power	$R = r,\ \ P = E^2/4r$	E, r fixed; R varied
Bridges	$P/Q = R/S$; $R/S = l/(100 - l)$	$I_g = 0$; no end error
Potentiometer	$E_1/E_2 = l_1/l_2$	uniform wire, $E_0 > E$
Power	$I^2R,\ \ V^2/R$	a resistor (VI: any device)

30 Common Traps

- Stretched n times: $R \to n^2R$, not nR.
- v_d is tiny, but the bulb lights at once.
- Max power ($R = r$) is not max efficiency (50%).
- In series (same rated V, R constant) the lower-wattage bulb is brighter.
- A real voltmeter on a discharging cell reads $E - Ir < E$; a balanced potentiometer draws no current and reads E.

31 Constants & Units

Quantity	Value
e	1.6×10^{-19} C
n (copper)	8.5×10^{28} m^{-3}
ρ (copper, 20°C)	$1.7 \times 10^{-8}\ \Omega$ m
μ ; σ	m^{2}V^{-1}s^{-1} ; S m^{-1}
1 kWh	3.6×10^6 J